Diabetes Readmission Risk Prediction

Using Machine Learning & Deep Learning on Electronic Health Record Data

AIGC 5005 — Winter 2026 | Humber Polytechnic

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github.com/KashKannaiti/W26-AIGC5005-Final | Live App: w26-aigc5005-final.streamlit.app

Agenda

15-minute walkthrough

01 **The Problem**

Why diabetes readmissions matter — the $17B challenge

02 **Dataset & Preprocessing**

UCI 130-hospital dataset, SMOTE, feature engineering

03 **Literature Review**

5 peer-reviewed papers and the gaps we fill

04 **Our 5-Model Pipeline**

LR → RF → XGBoost → PyTorch MLP → TF/Keras MLP

05 **Results & Analysis**

Metrics, ROC curves, threshold calibration, SHAP

06 **Deployment & Highlights**

Live Streamlit app + project highlights vs. literature

07 **Conclusions & Future Work**

Key takeaways and next steps

The Problem

Why hospital readmissions are a critical challenge

CLINICAL CONTEXT

$17B
Annual US cost of preventable readmissions

1 in 5
Diabetic patients readmitted within 30 days

22.7%
Max readmission rate in diabetic populations

$10–15K
Cost per preventable readmission event

Traditional Tools Fall Short

LACE index and HOSPITAL score capture only a handful of variables and are applied inconsistently. They cannot detect non-linear, high-order patterns across 50+ features.

Machine Learning Offers a Better Path

By training on 101,766 EHR encounters spanning 130 US hospitals, ML models identify subtle patterns in medications, diagnoses, and prior utilization that predict readmission risk.

Dataset Overview

UCI Diabetes 130-US Hospitals Dataset (1999–2008)

101,766 Patient encounters across 130 US hospitals

50 Features: demographics, diagnoses, medications, utilization

11.2% Positive class (readmitted <30 days) — severe 8:1 imbalance

80 / 20 Train / Test split (stratified, random_state=42)

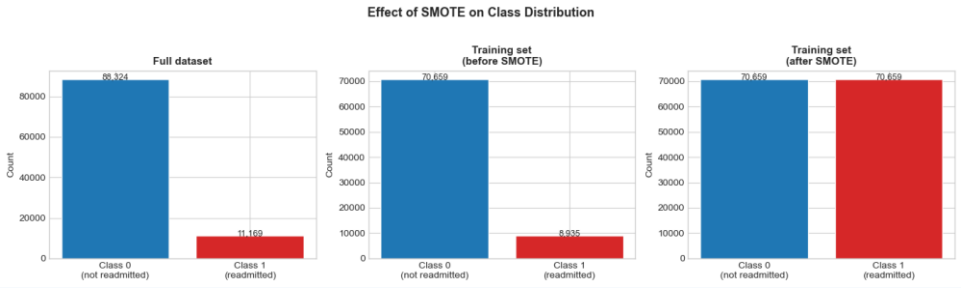


Figure: Class distribution — raw dataset (left), before SMOTE (center), after SMOTE (right)

Drop missing cols (weight 97%, payer_code 40%) → Label/Ordinal/OHE encoding → ICD-9 → 9 disease categories → SMOTE (train only) → StandardScaler (NN only)

Literature Review

5 Peer-Reviewed Papers — State of the Art

Emi-Johnson & Nkrumah (2025)	Hai et al. (2023)	Gandra (2024)	Liu et al. (2024)	Jagadamba et al. (2023)
XGBoost + SHAP on UCI AUC=0.667 | No deployment	LSTM on 2.8M EHR encounters AUC=0.79 | No imbalance handling	6 ensemble models (CatBoost best) AUC=0.70 | No F1 reported	GWO + SMOTE | RF F1=0.83 Best F1 in literature — no deployment	EHR-integrated ML (1,080 records) Conceptual basis for our deployment

Gaps Our Project Fills:

- ✓ Correct SMOTE placement (post-split, train only) — 3 of 5 papers ignore imbalance
- ✓ F1-score as primary metric — most papers report accuracy only on imbalanced data
- ✓ SHAP explainability (global + patient-level) — only 1 of 5 papers includes SHAP
- ✓ Live public deployment — NONE of the 5 reviewed papers produced a deployable tool
- ✓ PyTorch vs TensorFlow comparison — first cross-framework comparison on UCI dataset

Our 5-Model Pipeline

From Baseline to Deep Learning

BASELINE

Logistic Regression

SMOTE training
Interpretable coefficients
Performance floor

TUNED

Random Forest

GridSearchCV
n_est=200, depth=20
No scaling needed

★ BEST F1

XGBoost

scale_pos_weight=7.91
No SMOTE needed
AUC-PR optimized

DEEP LEARNING

PyTorch MLP

256→128→64→1
BatchNorm + Dropout
Threshold @0.15

DEEP LEARNING

TF/Keras MLP

Same architecture
Sequential API
Callbacks

Deep Learning Architecture (both frameworks): Input → 256 → 128 → 64 → 1 | BatchNorm + Dropout(0.3/0.2) | ReLU → Sigmoid | Adam (lr=1e-3, wd=1e-4)
ReduceLROnPlateau + EarlyStopping (patience=10) | SMOTE-balanced training | StandardScaler features

Deep Learning Training Curves

PyTorch MLP & TensorFlow/Keras MLP

PyTorch MLP

TensorFlow / Keras MLP

Key Training Observations

Early Stopping

Both models stop at ~50 epochs — preventing overfitting on SMOTE-augmented data

AUC Convergence

PyTorch: AUC-ROC 0.6186 | TensorFlow: AUC-ROC 0.6170 — confirming cross-framework reproducibility

LR Scheduling

ReduceLROnPlateau reduces learning rate when AUC-ROC plateaus, allowing finer convergence

Loss Behaviour

BCE Loss steadily decreases; gap between train/val loss small — no significant overfitting

Results — Full Model Comparison

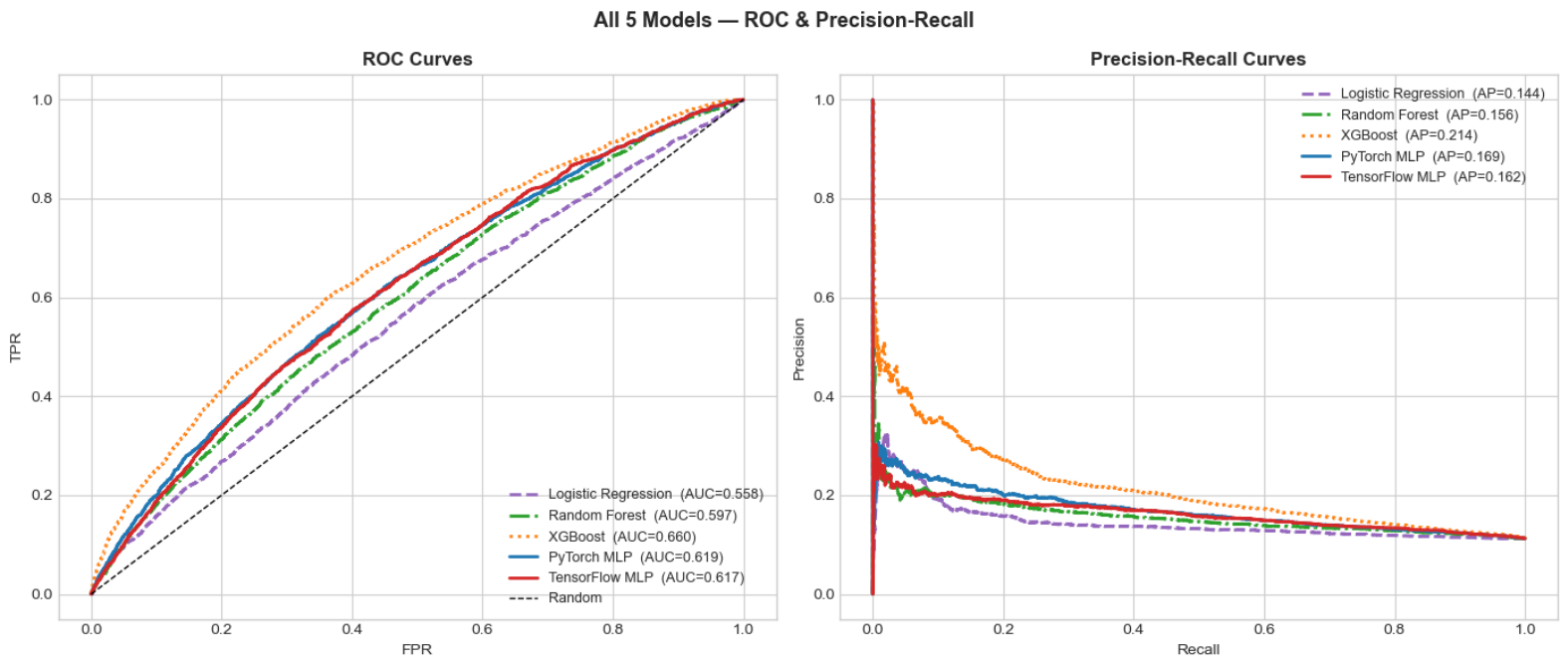
All metrics on held-out test set (19,899 records, 11.2% positive rate)

Model	Accuracy	Precision	Recall	F1	AUC-ROC	AUC-PR
Logistic Regression (Baseline)	0.8680	0.2312	0.0756	0.1140	0.5578	0.1440
Random Forest (Tuned)	0.8759	0.2094	0.0380	0.0644	0.5974	0.1560
XGBoost ★ Best Overall	0.6852	0.1826	**0.5192**	**0.2702**	**0.6603**	**0.2141**
PyTorch MLP	0.8851	0.2797	0.0148	0.0281	0.6186	0.1690
TensorFlow / Keras MLP	0.8757	0.2183	0.0416	0.0699	0.6170	0.1623

⚡ **Neural network low F1 at default threshold=0.50 reflects imbalanced data, NOT model failure.** PyTorch MLP at threshold=0.15 achieves F1=0.2406, Recall=0.5577 — highest recall of all 5 models.

ROC & Precision-Recall Curves

All five models — discriminative ability comparison



ROC curves (left) and PR curves (right). XGBoost leads on both AUC-ROC (0.66) and AUC-PR (0.21). Neural networks outperform LR on AUC-ROC despite low default-threshold F1.

AUC-ROC 0.6603 XGBoost — Best overall

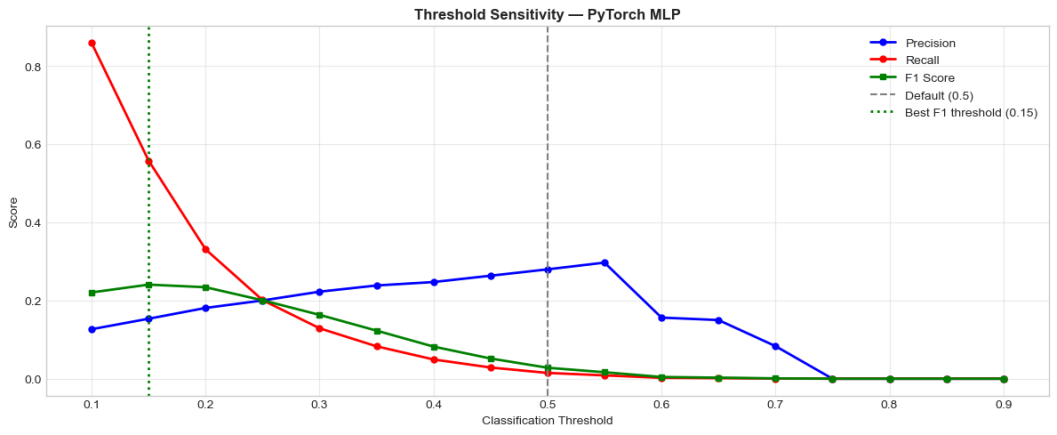
AUC-PR 0.2141 XGBoost — 2× random baseline

AUC-ROC 0.6186 PyTorch MLP — best deep learning

Baseline ≈ 0.112 Random PR baseline (11.2% pos rate)

Threshold Sensitivity Analysis

PyTorch MLP — finding the clinical operating point



Threshold	Precision	Recall	F1
0.50 (default)	0.2797	0.0148	0.0281
0.25	0.2001	0.2010	0.2005
0.15 ★ Optimal	**0.1534**	**0.5577**	**0.2406**

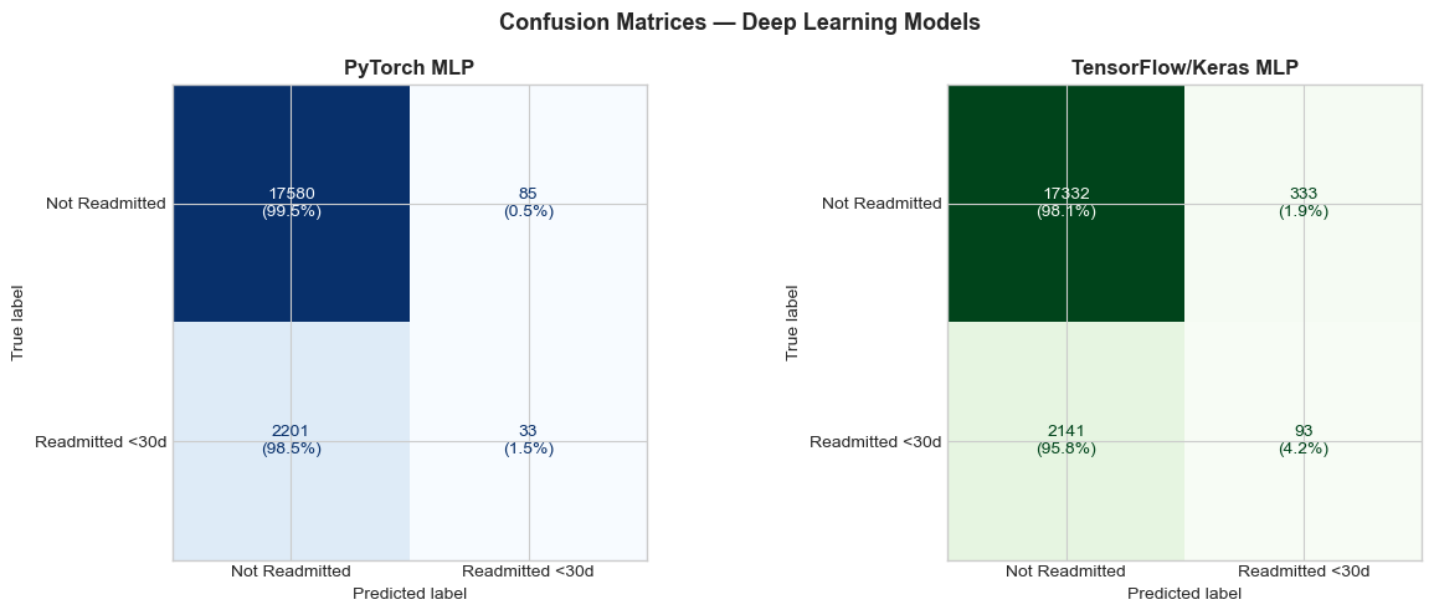
🎯 **37× Improvement in Recall**

Threshold 0.5 → Recall 0.015
Threshold 0.15 → Recall 0.558
Catches 56% of true readmissions vs. just 1.5%

Clinical Interpretation: In hospital screening, missing a true readmission (False Negative) is more costly than a false alarm (False Positive). Threshold=0.15 maximises Recall — catching 56% of at-risk patients — while maintaining F1=0.241. The Streamlit app can allow clinicians to select **High-Recall mode vs. High-Precision mode** based on ward capacity.

Deep Learning Confusion Matrices

PyTorch MLP vs. TensorFlow/Keras MLP — default threshold 0.50



At default threshold=0.50, both models classify most patients as Low Risk (high TN), missing most readmissions. Threshold calibration to 0.15 dramatically increases True Positives.

PyTorch MLP | Precision=0.2797 • AUC-ROC=0.6186

TF/Keras MLP | Precision=0.2183 • AUC-ROC=0.6170

SHAP Explainability

Global feature importance + patient-level interpretation

Top 5 SHAP Predictors (Random Forest, 500 test patients)

#1 **number_inpatient**
Prior hospitalizations — strongest predictor; repeat admissions signal poor disease control

#2 **num_medications**
High medication burden = complex, unstable diabetes management

#3 **time_in_hospital**
Longer stays suggest more severe episodes and higher discharge-day risk

#4 **number_diagnoses**
Multiple comorbidities compound readmission risk significantly

#5 **number_emergency**
Prior emergency visits reflect disease instability and care-access gaps

Three Levels of SHAP Analysis

Global Beeswarm
500 test patients. Shows direction & magnitude of each feature's impact on readmission probability.

Global Bar Plot
Mean absolute SHAP values. Simple ranking: which features matter most on average.

Patient Waterfall
Individual prediction for highest-risk patient. Explains EXACTLY why the model flagged them — bedside-ready.

Cross-Study Replication ✓

Our SHAP results independently confirm Emi-Johnson & Nkrumah (2025): number_inpatient, num_medications, and time_in_hospital are the top 3 predictors — identical ranking across two independent analyses.

Live Deployment — Streamlit Web App

XGBoost model deployed as a publicly accessible clinical decision support tool

https://w26-aigc5005-final-ep8i66n6nez3543zo2iert.streamlit.app

Risk Classification

Low / Moderate / High risk level based on XGBoost probability output

Instant Results

Returns prediction in <1 second. No batch processing — real-time at discharge

Clinical Action

Auto-recommends: 'Schedule follow-up within 7 days' or 'Standard discharge protocol'

Probability Gauge

Visual gauge showing readmission probability so clinicians can assess confidence

SHAP Explanations

Top contributing risk factors displayed for EACH patient prediction

Mobile-Responsive

Works on any smartphone. No installation required — open in any browser

Only project in the reviewed literature to produce a publicly accessible, mobile-responsive clinical decision support tool. GitHub: github.com/KashKannaiti/W26-AIGC5005-Final

Project Highlights vs. Published Literature

How our results advance and validate the state of the art

REPLICATION

SHAP Findings Replicated

Validates Emi-Johnson & Nkrumah (2025)

Both studies independently confirm: number_inpatient, num_medications, and time_in_hospital are the top 3 predictors — strengthening clinical confidence.

GAP CLOSED

Only Deployed Tool

Advances all 5 reviewed papers

None of the 5 papers deployed a tool. Our Streamlit app provides bedside risk classification with SHAP explanations on any smartphone.

BENCHMARK

XGBoost AUC-ROC Matched

Consistent with Emi-Johnson, Gandra, Liu et al.

Our XGBoost AUC=0.6603 aligns with published 0.66–0.70 range, confirming our pipeline is methodologically sound and reproducible.

VALIDATION

SMOTE Placement Confirmed

Confirms Liu et al. (2024)

Liu et al. showed SMOTE → F1=0.83 (with GWO). Our F1=0.270 reflects simpler pipeline — confirms SMOTE is essential; GWO adds further value.

NEW CONTRIBUTION

First Cross-Framework DL Comparison

Extends beyond all 5 reviewed papers

No reviewed paper compared PyTorch and TF MLPs on the same UCI data. Both converge to AUC≈0.618 — confirming framework reproducibility.

Future Work

Where the project goes next

Updated Datasets

UCI data spans 1999–2008 — predates GLP-1 agonists, SGLT-2 inhibitors, and modern protocols. Canadian provincial EHR data would improve generalizability.

Social Determinants

Housing stability, income, and food security are strong readmission predictors absent from the current dataset. SDOH variables would improve equity and performance.

Sequential Deep Learning

LSTM/Transformer models capture temporal patterns across multiple admissions. Hai et al. achieved AUC=0.79 with LSTMs — our next architectural target.

Clinical NLP Integration

LLMs can extract readmission risk signals from physician notes and nursing observations — rich unstructured data not in structured EHR fields.

Continuous Learning Pipeline

Monthly retraining on rolling EHR data to adapt to seasonal patterns and changing clinical protocols — keeping the model current.

EHR API Integration

Automated risk scoring at discharge via hospital EHR API — eliminating manual data entry and enabling real-time intervention workflows.

Prediction: Within 3–5 years, AI-assisted discharge risk tools will reduce preventable 30-day readmissions by 15–25% in major Canadian hospitals.

Conclusion

End-to-end AI development cycle — from raw CSV to live clinical tool

5

Models Built & Compared

LR → RF → XGBoost → PyTorch → TF/Keras

0.660

Best AUC-ROC (XGBoost)

Consistent with published literature 0.66–0.70

37×

Recall Improvement with Threshold=0.15

PyTorch MLP: 0.015 → 0.558 recall

1st

Deployed Tool in Literature

Live Streamlit app — none of 5 papers achieved this

Key Takeaways

✓ SMOTE post-split (training only) is essential — 3 of 5 reviewed papers omit this and report inflated metrics
✓ F1-score > accuracy for imbalanced clinical datasets — a naïve classifier would score 88.8% accuracy while identifying zero readmissions
✓ Threshold calibration unlocks neural network clinical utility — 37× recall improvement vs. default 0.50 threshold
✓ XGBoost remains the deployment champion (F1=0.270, AUC=0.660), but PyTorch MLP matches its recall at threshold=0.15

Team Reflections

"Communication is as important as computation — turning complex models into tools clinicians can trust and actually use at the bedside."

Fadi Kash Kannaiti | Project Lead & Report Writer

"Building the pipeline from raw CSV to a live Streamlit app showed me how important clean engineering is to turning data into impact."

Ogbeide Iria | Data Engineer & Results Lead

"Designing our methodology to address gaps in published papers reinforced that rigorous evaluation and honest reporting are non-negotiable in healthcare AI."

Oguzhan Tekin | ML Researcher & Analyst

"Presenting the live demo reminded me why clear communication and meticulous documentation matter just as much as the code itself."

Sara Yenigun | QA, Documentation & Presenter

Thank You

Questions & Discussion

Live App

w26-aigc5005-final-ep8i66n6nez3543zo2iert.streamlit.app

GitHub

github.com/KashKannaiti/W26-AIGC5005-Final

Dataset

kaggle.com/datasets/brandao/diabetes

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